ABHISHEK SAHOO DATA CLEANING IN PYTHON

```
# Importing Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
# Importing Dataset
df = pd.read_csv("messy_customer_sales_data.csv")
df
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	m	52.0	KOLKATA	2025-06-26	2025-05-17	26944.0	1.0
1	CUST5957	Jason Thomas	M	51.0 years	NaN	2021-02-17	2025-07-22	44152.0	2.0
2	CUST3754	Brittney Martinez	F	62.0	hyderabad	2023-11-05	2024-12-08	31745.0	2.0
3	CUST2934	Brenda Pierce	FEMALE	40.0	hyderabad	2022-03-13	2025-10-02	39674.0	1.0
4	CUST5683	Matthew Carroll	f	41.0	CHENNAI	2024-04-05	2024-12-15	NaN	8.0
...	...	...	...	...	...	...	...	...	...
10195	CUST10767	Robert Lewis	female	35.0 years	delhi	2020-12-08	2025-01-25	24167.0	9.0
10196	NaN	Diane Evans	M	53.0	bangalore	2023-12-31	2025-05-07	11639.0	7.0
10197	CUST6315	Joshua Martinez	m	25.0	hyderabad	2022-02-15	2025-01-11	43832.0	2.0
10198	CUST4812	Sarah Miller	FEMALE	55.0	NaN	2021-03-16	2025-05-14	18827.0	10.0
10199	CUST6588	David Potter	female	34.0	HYDERABAD	2020-10-12	2025-06-27	35211.0	10.0

10200 rows × 12 columns

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	m	52.0	KOLKATA	2025-06-26	2025-05-17	26944.0	1.0
1	CUST5957	Jason Thomas	M	51.0 years	NaN	2021-02-17	2025-07-22	44152.0	2.0
2	CUST3754	Brittney Martinez	F	62.0	hyderabad	2023-11-05	2024-12-08	31745.0	2.0
3	CUST2934	Brenda Pierce	FEMALE	40.0	hyderabad	2022-03-13	2025-10-02	39674.0	1.0
4	CUST5683	Matthew Carroll	f	41.0	CHENNAI	2024-04-05	2024-12-15	NaN	8.0
...	...	...	...	...	...	...	...	...	...
10195	CUST10767	Robert Lewis	female	35.0 years	delhi	2020-12-08	2025-01-25	24167.0	9.0
10196	NaN	Diane Evans	M	53.0	bangalore	2023-12-31	2025-05-07	11639.0	7.0
10197	CUST6315	Joshua Martinez	m	25.0	hyderabad	2022-02-15	2025-01-11	43832.0	2.0
10198	CUST4812	Sarah Miller	FEMALE	55.0	NaN	2021-03-16	2025-05-14	18827.0	10.0
10199	CUST6588	David Potter	female	34.0	HYDERABAD	2020-10-12	2025-06-27	35211.0	10.0

10200 rows × 12 columns

Next steps: Generate code with df | New interactive sheet | Generate code with df | New interactive sheet

```
# Checking Null Values
df.isnull().sum()
```

	0
Customer_ID	1023
Name	0
Gender	1026
Age	951
City	1016
Signup_Date	0
Last_Purchase_Date	1012
Purchase_Amount	1021
Feedback_Score	1023
Email	0
Phone_Number	0
Country	732

dtype: int64

	0
Customer_ID	1023
Name	0
Gender	1026
Age	951
City	1016
Signup_Date	0
Last_Purchase_Date	1012
Purchase_Amount	1021
Feedback_Score	1023
Email	0
Phone_Number	0
Country	732

dtype: int64


```
# Dropping Null values in the Customer_ID column
df.dropna(subset = ['Customer_ID'], inplace = True)
```

```
df.isnull().sum()
```

	0
Customer_ID	0
Name	0
Gender	934
Age	859
City	918
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	927
Feedback_Score	905
Email	0
Phone_Number	0
Country	664

dtype: int64

	0
Customer_ID	0
Name	0
Gender	934
Age	859
City	918
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	927
Feedback_Score	905
Email	0
Phone_Number	0
Country	664

dtype: int64


```
# Transformation of Age column
# Checking Unique values in Age column
df['Age'].unique()
```

```
array(['52.0', '51.0 years', '62.0', '40.0', '41.0', nan, '18.0',
       '43.0 years', '40.0 years', '26.0', '32.0', '22.0', '59.0', '65.0',
       '61.0', '31.0', '54.0 years', '55.0', '69.0', '61.0 years', '24.0',
       '63.0', '19.0', '50.0', '56.0', '36.0', '68.0', '43.0', '38.0',
       '27.0', '57.0 years', '23.0', '25.0', '66.0', '28.0', '30.0',
       '46.0', '48.0', '20.0', '37.0', '67.0', '51.0', '35.0', '58.0',
       '29.0', 'nan years', '39.0', '49.0', '47.0', '42.0', '44.0',
       '64.0', '53.0', '60.0', '59.0 years', '45.0', '21.0', '34.0',
       '54.0', '48.0 years', '46.0 years', '33.0', '57.0', '30.0 years',
       '58.0 years', '35.0 years', '34.0 years', '69.0 years', '250',
       '19.0 years', '27.0 years', '53.0 years', '65.0 years',
       '66.0 years', '44.0 years', '49.0 years', '25.0 years',
       '23.0 years', '62.0 years', '41.0 years', '33.0 years',
       '28.0 years', '22.0 years', '20.0 years', '42.0 years',
       '45.0 years', '3', '63.0 years', '37.0 years', '38.0 years',
       '55.0 years', '18.0 years', '36.0 years', '67.0 years',
       '29.0 years', '39.0 years', '31.0 years', '47.0 years',
       '52.0 years', '-10', '60.0 years', '24.0 years', '26.0 years',
       '21.0 years', '64.0 years', '50.0 years', '68.0 years',
       '32.0 years', '56.0 years'], dtype=object)
```


```
array(['52.0', '51.0 years', '62.0', '40.0', '41.0', nan, '18.0',
       '43.0 years', '40.0 years', '26.0', '32.0', '22.0', '59.0', '65.0',
       '61.0', '31.0', '54.0 years', '55.0', '69.0', '61.0 years', '24.0',
       '63.0', '19.0', '50.0', '56.0', '36.0', '68.0', '43.0', '38.0',
       '27.0', '57.0 years', '23.0', '25.0', '66.0', '28.0', '30.0',
       '46.0', '48.0', '20.0', '37.0', '67.0', '51.0', '35.0', '58.0',
       '29.0', 'nan years', '39.0', '49.0', '47.0', '42.0', '44.0',
```

```
       '64.0', '53.0', '60.0', '59.0 years', '45.0', '21.0', '34.0',
       '54.0', '48.0 years', '46.0 years', '33.0', '57.0', '30.0 years',
       '58.0 years', '35.0 years', '34.0 years', '69.0 years', '250',
       '19.0 years', '27.0 years', '53.0 years', '65.0 years',
       '66.0 years', '44.0 years', '49.0 years', '25.0 years',
       '23.0 years', '62.0 years', '41.0 years', '33.0 years',
       '28.0 years', '22.0 years', '20.0 years', '42.0 years',
       '45.0 years', '3', '63.0 years', '37.0 years', '38.0 years',
       '55.0 years', '18.0 years', '36.0 years', '67.0 years',
       '29.0 years', '39.0 years', '31.0 years', '47.0 years',
       '52.0 years', '-10', '60.0 years', '24.0 years', '26.0 years',
       '21.0 years', '64.0 years', '50.0 years', '68.0 years',
       '32.0 years', '56.0 years'], dtype=object)
```

```
# Replacing the 'years' to blanks in Age column
df['Age'] = df['Age'].str.replace('years', '')
```

```
df['Age'].unique()
```

```
array(['52.0', '51.0 ', '62.0', '40.0', '41.0', nan, '18.0', '43.0 ',
       '40.0 ', '26.0', '32.0', '22.0', '59.0', '65.0', '61.0', '31.0',
       '54.0 ', '55.0', '69.0', '61.0 ', '24.0', '63.0', '19.0', '50.0',
       '56.0', '36.0', '68.0', '43.0', '38.0', '27.0', '57.0 ', '23.0',
       '25.0', '66.0', '28.0', '30.0', '46.0', '48.0', '20.0', '37.0',
       '67.0', '51.0', '35.0', '58.0', '29.0', 'nan ', '39.0', '49.0',
       '47.0', '42.0', '44.0', '64.0', '53.0', '60.0', '59.0 ', '45.0',
       '21.0', '34.0', '54.0', '48.0 ', '46.0 ', '33.0', '57.0', '30.0 ',
       '58.0 ', '35.0 ', '34.0 ', '69.0 ', '250', '19.0 ', '27.0 ',
       '53.0 ', '65.0 ', '66.0 ', '44.0 ', '49.0 ', '25.0 ', '23.0 ',
       '62.0 ', '41.0 ', '33.0 ', '28.0 ', '22.0 ', '20.0 ', '42.0 ',
       '45.0 ', '3', '63.0 ', '37.0 ', '38.0 ', '55.0 ', '18.0 ', '36.0 ',
       '67.0 ', '29.0 ', '39.0 ', '31.0 ', '47.0 ', '52.0 ', '-10',
       '60.0 ', '24.0 ', '26.0 ', '21.0 ', '64.0 ', '50.0 ', '68.0 ',
       '32.0 ', '56.0 '], dtype=object)
```

```
array(['52.0', '51.0 ', '62.0', '40.0', '41.0', nan, '18.0', '43.0 ',
       '40.0 ', '26.0', '32.0', '22.0', '59.0', '65.0', '61.0', '31.0',
       '54.0 ', '55.0', '69.0', '61.0 ', '24.0', '63.0', '19.0', '50.0',
       '56.0', '36.0', '68.0', '43.0', '38.0', '27.0', '57.0 ', '23.0',
       '25.0', '66.0', '28.0', '30.0', '46.0', '48.0', '20.0', '37.0',
       '67.0', '51.0', '35.0', '58.0', '29.0', 'nan ', '39.0', '49.0',
       '47.0', '42.0', '44.0', '64.0', '53.0', '60.0', '59.0 ', '45.0',
       '21.0', '34.0', '54.0', '48.0 ', '46.0 ', '33.0', '57.0', '30.0 ',
       '58.0 ', '35.0 ', '34.0 ', '69.0 ', '250', '19.0 ', '27.0 ',
       '53.0 ', '65.0 ', '66.0 ', '44.0 ', '49.0 ', '25.0 ', '23.0 ',
       '62.0 ', '41.0 ', '33.0 ', '28.0 ', '22.0 ', '20.0 ', '42.0 ',
       '45.0 ', '3', '63.0 ', '37.0 ', '38.0 ', '55.0 ', '18.0 ', '36.0 ',
       '67.0 ', '29.0 ', '39.0 ', '31.0 ', '47.0 ', '52.0 ', '-10',
       '60.0 ', '24.0 ', '26.0 ', '21.0 ', '64.0 ', '50.0 ', '68.0 ',
       '32.0 ', '56.0 '], dtype=object)
```


```
# Removing the whitespaces / unnecessary spaces in Age column
df['Age'] = df['Age'].str.strip()
```

```
df['Age'] = df['Age'].str.strip('.0')
```

```
df['Age'].unique()
```

```
array(['52', '51', '62', '4', '41', nan, '18', '43', '26', '32', '22',
       '59', '65', '61', '31', '54', '55', '69', '24', '63', '19', '5',
       '56', '36', '68', '38', '27', '57', '23', '25', '66', '28', '3',
       '46', '48', '2', '37', '67', '35', '58', '29', 'nan', '39', '49',
       '47', '42', '44', '64', '53', '6', '45', '21', '34', '33', '-1'],
      dtype=object)
```

```
array(['52', '51', '62', '4', '41', nan, '18', '43', '26', '32', '22',
       '59', '65', '61', '31', '54', '55', '69', '24', '63', '19', '5',
       '56', '36', '68', '38', '27', '57', '23', '25', '66', '28', '3',
       '46', '48', '2', '37', '67', '35', '58', '29', 'nan', '39', '49',
       '47', '42', '44', '64', '53', '6', '45', '21', '34', '33', '-1'],
      dtype=object)
```


```
# Changing the Age column into int / float ( Here we cannot use astype method because we have null values and theref
df['Age'] = pd.to_numeric(df['Age'], errors = 'coerce')
```

```
df['Age'].median()
```

```
42.0
```

```
42.0
```

```
# Replacing the Null values in Age column to Median value
df['Age'].fillna(df['Age'].median(), inplace = True)
```

```
/tmp/ipykernel_3175/1112563432.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Age'].fillna(df['Age'].median(), inplace = True)
/tmp/ipykernel_3175/1112563432.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Age'].fillna(df['Age'].median(), inplace = True)
```

```
df.isnull().sum()
```

	0
Customer_ID	0
Name	0
Gender	934
Age	0
City	918
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	927
Feedback_Score	905
Email	0
Phone_Number	0
Country	664

dtype: int64

	0
Customer_ID	0
Name	0
Gender	934
Age	0
City	918
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	927
Feedback_Score	905
Email	0
Phone_Number	0
Country	664

dtype: int64


```
df[df['Age'] == -1]
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score	
2702	CUST1941	Virginia Casey	male	-1.0	chennai	2022-09-22	2025-01-12	3056.0	2.0	r
7720	CUST3810	Meredith Shaw	M	-1.0	KOLKATA	2024-05-21	2025-07-03	6126.0	1.0	
8337	CUST8792	Mark Meyer	female	-1.0	delhi	2022-03-29	2025-04-09	22740.0	1.0	tyleı

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score	
2702	CUST1941	Virginia Casey	male	-1.0	chennai	2022-09-22	2025-01-12	3056.0	2.0	r
7720	CUST3810	Meredith Shaw	M	-1.0	KOLKATA	2024-05-21	2025-07-03	6126.0	1.0	
8337	CUST8792	Mark Meyer	female	-1.0	delhi	2022-03-29	2025-04-09	22740.0	1.0	tyleı


```
df['Age'].dtype
```

```
dtype('float64')
```

```
dtype('float64')
```


```
df['Age'] = df['Age'].astype('int64')
```

```
df['Age'].dtype
```

```
dtype('int64')
```

```
dtype('int64')
```


```
# Replacing the Null values to Median value
df['Purchase_Amount'].fillna(df['Purchase_Amount'].median(), inplace = True)
```

```
/tmp/ipykernel_3175/2388332072.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Purchase_Amount'].fillna(df['Purchase_Amount'].median(), inplace = True)
```

```
/tmp/ipykernel_3175/2388332072.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Purchase_Amount'].fillna(df['Purchase_Amount'].median(), inplace = True)
```


```
# Replacing the Null values with Mode Value
df['Feedback_Score'].fillna(df['Feedback_Score'].mode()[0], inplace = True)
```

```
/tmp/ipykernel_3175/3016619845.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Feedback_Score'].fillna(df['Feedback_Score'].mode()[0], inplace = True)
```

```
/tmp/ipykernel_3175/3016619845.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series th
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df['Feedback_Score'].fillna(df['Feedback_Score'].mode()[0], inplace = True)
```


```
# Replacing the Null values Gender, City, Country column with Mode value
for col in ['Gender', 'City', 'Country']:
  df[col].fillna(df[col].mode()[0],inplace = True)
```

```
/tmp/ipykernel_3175/927364277.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series thr
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which
```

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df[col].fillna(df[col].mode()[0],inplace = True)
/tmp/ipykernel_3175/927364277.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series thr
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d


  df[col].fillna(df[col].mode()[0],inplace = True)
```

```
df.isnull().sum()
```

	0
Customer_ID	0
Name	0
Gender	0
Age	0
City	0
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	0
Feedback_Score	0
Email	0
Phone_Number	0
Country	0

dtype: int64

	0
Customer_ID	0
Name	0
Gender	0
Age	0
City	0
Signup_Date	0
Last_Purchase_Date	914
Purchase_Amount	0
Feedback_Score	0
Email	0
Phone_Number	0
Country	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(14)
```
np.int64(14)

```
df = df.dropna()
```

```
df.shape
```

```
(8263, 12)
```
(8263, 12)

```
df['Gender'].unique()
```

```
array(['m ', 'M', 'F', 'FEMALE', 'f ', 'male', 'MALE', 'female'],
      dtype=object)
array(['m ', 'M', 'F', 'FEMALE', 'f ', 'male', 'MALE', 'female'],
      dtype=object)
```

```
df['Gender'] = df['Gender'].str.strip()
```

```
/tmp/ipykernel_3175/64578066.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.strip()
/tmp/ipykernel_3175/64578066.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.strip()
```

```
df['Gender'].unique()
```

```
array(['m', 'M', 'F', 'FEMALE', 'f', 'male', 'MALE', 'female'],
      dtype=object)
array(['m', 'M', 'F', 'FEMALE', 'f', 'male', 'MALE', 'female'],
      dtype=object)
```

```
df['Gender'] = df['Gender'].str.replace('F', 'Female')
df['Gender'] = df['Gender'].str.replace('M', 'Male')
df['Gender'] = df['Gender'].str.replace('male', 'Male')
df['Gender'] = df['Gender'].str.replace('MALE', 'Male')
df['Gender'] = df['Gender'].str.replace('female', 'Female')
df['Gender'] = df['Gender'].str.replace('m', 'Male')
df['Gender'] = df['Gender'].str.replace('f', 'Female')
df['Gender'] = df['Gender'].str.replace('FEMALE', 'Female')
```

```
  df['Gender'] = df['Gender'].str.replace('female', 'Female')
/tmp/ipykernel_3175/430723032.py:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returnir
  df['Gender'] = df['Gender'].str.replace('m', 'Male')
/tmp/ipykernel_3175/430723032.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returnir
  df['Gender'] = df['Gender'].str.replace('f', 'Female')
/tmp/ipykernel_3175/430723032.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returnir
  df['Gender'] = df['Gender'].str.replace('FEMALE', 'Female')
```

```
df['Gender'].unique()
```

```
array(['Male', 'FeMale', 'FeMaleEMaleALE', 'Female', 'MaleALE',
       'FemaleeMale'], dtype=object)
array(['Male', 'FeMale', 'FeMaleEMaleALE', 'Female', 'MaleALE',
       'FemaleeMale'], dtype=object)
```

```
df['Gender'] = df['Gender'].str.replace('FeMaleEMaleALE', 'Female')
df['Gender'] = df['Gender'].str.replace('MaleALE', 'Male')
df['Gender'] = df['Gender'].str.replace('FemaleeMale', 'Female')
df['Gender'] = df['Gender'].str.replace('FeMale', 'Female')
```

```
/tmp/ipykernel_3175/3889817714.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FeMaleEMaleALE', 'Female')
/tmp/ipykernel_3175/3889817714.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('MaleALE', 'Male')
/tmp/ipykernel_3175/3889817714.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FemaleeMale', 'Female')
/tmp/ipykernel_3175/3889817714.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FeMale', 'Female')
/tmp/ipykernel_3175/3889817714.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FeMaleEMaleALE', 'Female')
/tmp/ipykernel_3175/3889817714.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('MaleALE', 'Male')
/tmp/ipykernel_3175/3889817714.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FemaleeMale', 'Female')
/tmp/ipykernel_3175/3889817714.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Gender'] = df['Gender'].str.replace('FeMale', 'Female')
```

```
df['Gender'].unique()
```

```
array(['Male', 'Female'], dtype=object)
array(['Male', 'Female'], dtype=object)
```

```
df['City'].unique()
```

```
array([' KOLKATA ', ' Kolkata ', ' hyderabad ', ' CHENNAI ', ' kolkata ',
       ' BANGALORE ', ' Hyderabad ', ' HYDERABAD ', ' Mumbai ',
       ' chennai ', ' Delhi ', ' bangalore ', ' Bangalore ', ' delhi ',
       ' MUMBAI ', ' Chennai ', ' DELHI ', ' mumbai '], dtype=object)
array([' KOLKATA ', ' Kolkata ', ' hyderabad ', ' CHENNAI ', ' kolkata ',
       ' BANGALORE ', ' Hyderabad ', ' HYDERABAD ', ' Mumbai ',
       ' chennai ', ' Delhi ', ' bangalore ', ' Bangalore ', ' delhi ',
       ' MUMBAI ', ' Chennai ', ' DELHI ', ' mumbai '], dtype=object)
```

```
df['City'] = df['City'].str.strip().str.lower()
```

```
/tmp/ipykernel_3175/357960059.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['City'] = df['City'].str.strip().str.lower()
/tmp/ipykernel_3175/357960059.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['City'] = df['City'].str.strip().str.lower()
```

```
df['City'].unique()
```

```
array(['kolkata', 'hyderabad', 'chennai', 'bangalore', 'mumbai', 'delhi'],
      dtype=object)
array(['kolkata', 'hyderabad', 'chennai', 'bangalore', 'mumbai', 'delhi'],
      dtype=object)
```

```
df['Country'].unique()
```

```
array(['India', 'india', 'InDia', 'IND'], dtype=object)
array(['India', 'india', 'InDia', 'IND'], dtype=object)
```

```
df['Country'] = df['Country'].str.lower()
```

```
/tmp/ipykernel_3175/2268126652.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.lower()
/tmp/ipykernel_3175/2268126652.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.lower()
```

```
df['Country'].unique()
```

```
array(['india', 'ind'], dtype=object)
array(['india', 'ind'], dtype=object)
```

```
df['Country'] = df['Country'].str.replace('ind', 'india')
```

```
/tmp/ipykernel_3175/3260782105.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.replace('ind', 'india')
/tmp/ipykernel_3175/3260782105.py:1: SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.replace('ind', 'india')
```

```
df['Country'].unique()
```

```
array(['indiaia', 'india'], dtype=object)
```

array(['indiaia', 'india'], dtype=object)

```
df['Country'] = df['Country'].str.replace('indiaia', 'india')
```

```
/tmp/ipykernel_3175/2519039748.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.replace('indiaia', 'india')
```

/tmp/ipykernel_3175/2519039748.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Country'] = df['Country'].str.replace('indiaia', 'india')

```
df['Country'].unique()
```

```
array(['india'], dtype=object)
```

array(['india'], dtype=object)

```
df.duplicated().sum()
```

```
np.int64(163)
```

np.int64(163)

```
df.drop_duplicates()
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	kolkata	2025-06-26	2025-05-17	26944.0	1.0
1	CUST5957	Jason Thomas	Male	51	kolkata	2021-02-17	2025-07-22	44152.0	2.0
2	CUST3754	Brittney Martinez	Female	62	hyderabad	2023-11-05	2024-12-08	31745.0	2.0
3	CUST2934	Brenda Pierce	Female	4	hyderabad	2022-03-13	2025-10-02	39674.0	1.0
4	CUST5683	Matthew Carroll	Female	41	chennai	2024-04-05	2024-12-15	24268.0	8.0
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	kolkata	2023-03-04	2025-06-30	43655.0	6.0
10194	CUST6146	Cody Thompson	Female	4	kolkata	2024-08-21	2024-10-26	35101.0	1.0
10195	CUST10767	Robert Lewis	Female	35	delhi	2020-12-08	2025-01-25	24167.0	9.0
10197	CUST6315	Joshua Martinez	Male	25	hyderabad	2022-02-15	2025-01-11	43832.0	2.0
10198	CUST4812	Sarah Miller	Female	55	kolkata	2021-03-16	2025-05-14	18827.0	10.0

8100 rows × 12 columns

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	kolkata	2025-06-26	2025-05-17	26944.0	1.0
1	CUST5957	Jason Thomas	Male	51	kolkata	2021-02-17	2025-07-22	44152.0	2.0
2	CUST3754	Brittney Martinez	Female	62	hyderabad	2023-11-05	2024-12-08	31745.0	2.0
3	CUST2934	Brenda Pierce	Female	4	hyderabad	2022-03-13	2025-10-02	39674.0	1.0
4	CUST5683	Matthew Carroll	Female	41	chennai	2024-04-05	2024-12-15	24268.0	8.0
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	kolkata	2023-03-04	2025-06-30	43655.0	6.0
10194	CUST6146	Cody Thompson	Female	4	kolkata	2024-08-21	2024-10-26	35101.0	1.0
10195	CUST10767	Robert Lewis	Female	35	delhi	2020-12-08	2025-01-25	24167.0	9.0
10197	CUST6315	Joshua Martinez	Male	25	hyderabad	2022-02-15	2025-01-11	43832.0	2.0
10198	CUST4812	Sarah Miller	Female	55	kolkata	2021-03-16	2025-05-14	18827.0	10.0

8100 rows × 12 columns


```
df.shape
```

```
(8263, 12)
(8263, 12)
```

```
df.dtypes
```

	0
Customer_ID	object
Name	object
Gender	object
Age	int64
City	object
Signup_Date	object
Last_Purchase_Date	object
Purchase_Amount	float64
Feedback_Score	float64
Email	object
Phone_Number	int64
Country	object

dtype: object

	0
Customer_ID	object
Name	object
Gender	object
Age	int64
City	object
Signup_Date	object
Last_Purchase_Date	object
Purchase_Amount	float64
Feedback_Score	float64
Email	object
Phone_Number	int64
Country	object

dtype: object


```
# Changing the date column into datetime format
df['Signup_Date'] = pd.to_datetime(df['Signup_Date'])
```

```
/tmp/ipykernel_3175/3276389993.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Signup_Date'] = pd.to_datetime(df['Signup_Date'])
/tmp/ipykernel_3175/3276389993.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Signup_Date'] = pd.to_datetime(df['Signup_Date'])
```

```
# Changing the date column into datetime format
df['Last_Purchase_Date'] = pd.to_datetime(df['Last_Purchase_Date'])
```

```
/tmp/ipykernel_3175/3137903953.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Last_Purchase_Date'] = pd.to_datetime(df['Last_Purchase_Date'])
/tmp/ipykernel_3175/3137903953.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Last_Purchase_Date'] = pd.to_datetime(df['Last_Purchase_Date'])
```

```
df['Feedback_Score'] = df['Feedback_Score'].astype('int64')
```

```
/tmp/ipykernel_3175/668418748.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Feedback_Score'] = df['Feedback_Score'].astype('int64')
/tmp/ipykernel_3175/668418748.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df['Feedback_Score'] = df['Feedback_Score'].astype('int64')
```

```
df.dtypes
```

	0
Customer_ID	object
Name	object
Gender	object
Age	int64
City	object
Signup_Date	datetime64[ns]
Last_Purchase_Date	datetime64[ns]
Purchase_Amount	float64
Feedback_Score	int64
Email	object
Phone_Number	int64
Country	object

dtype: object

	0
Customer_ID	object
Name	object
Gender	object
Age	int64
City	object
Signup_Date	datetime64[ns]
Last_Purchase_Date	datetime64[ns]
Purchase_Amount	float64
Feedback_Score	int64
Email	object
Phone_Number	int64
Country	object

dtype: object


```
df.duplicated().sum()
```

```
np.int64(163)
np.int64(163)
```

```
df['Customer_ID'].value_counts()
```

	count
Customer_ID	
CUST5361	2
CUST2239	2
CUST10040	2
CUST1091	2
CUST1579	2
...	...
CUST5816	1
CUST1029	1
CUST3972	1
CUST10298	1
CUST4647	1

8097 rows × 1 columns

dtype: int64

	count
Customer_ID	
CUST5361	2
CUST2239	2
CUST10040	2
CUST1091	2
CUST1579	2
...	...
CUST5816	1
CUST1029	1
CUST3972	1
CUST10298	1
CUST4647	1

8097 rows × 1 columns

dtype: int64


```
# Removing the duplicate rows from Customer ID by keeping the first record
df.drop_duplicates(subset = ['Customer_ID'], keep = 'first', inplace = True)
```

```
/tmp/ipykernel_3175/3229402727.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df.drop_duplicates(subset = ['Customer_ID'], keep = 'first', inplace = True)
/tmp/ipykernel_3175/3229402727.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
  df.drop_duplicates(subset = ['Customer_ID'], keep = 'first', inplace = True)
```

```
df['Customer_ID'].value_counts()
```

	count
Customer_ID	
CUST4812	1
CUST9038	1
CUST1565	1
CUST3881	1
CUST7857	1
...	...
CUST1516	1
CUST5683	1
CUST2934	1
CUST3754	1
CUST5957	1

8097 rows × 1 columns

dtype: int64

	count
Customer_ID	
CUST4812	1
CUST9038	1
CUST1565	1
CUST3881	1
CUST7857	1
...	...
CUST1516	1
CUST5683	1
CUST2934	1
CUST3754	1
CUST5957	1

8097 rows × 1 columns

dtype: int64


```
df.duplicated().sum()
```

```
np.int64(0)
np.int64(0)
```

```
df.isnull().sum()
```

	0
Customer_ID	0
Name	0
Gender	0
Age	0
City	0
Signup_Date	0
Last_Purchase_Date	0
Purchase_Amount	0
Feedback_Score	0
Email	0
Phone_Number	0
Country	0

dtype: int64

	0
Customer_ID	0
Name	0
Gender	0
Age	0
City	0
Signup_Date	0
Last_Purchase_Date	0
Purchase_Amount	0
Feedback_Score	0
Email	0
Phone_Number	0
Country	0

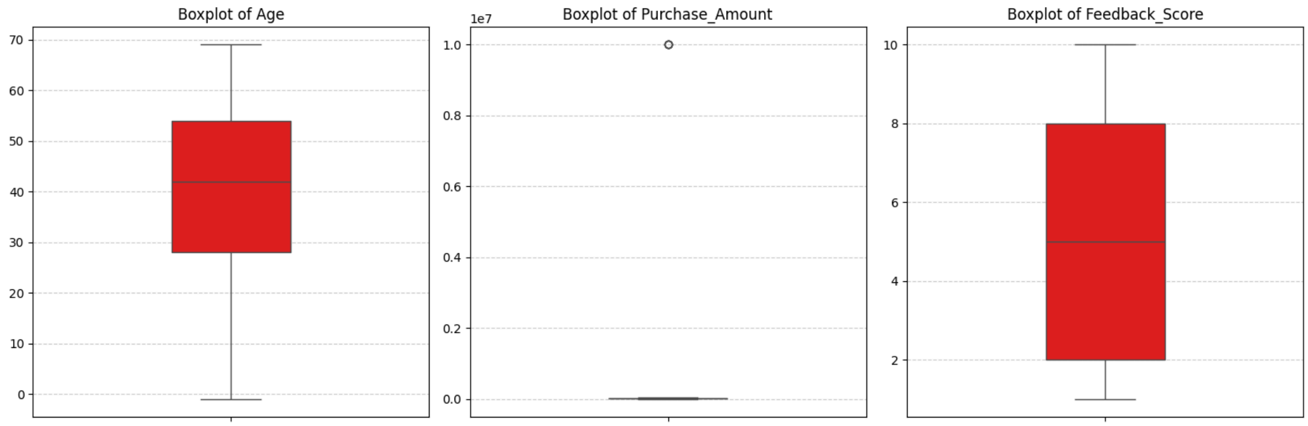
dtype: int64

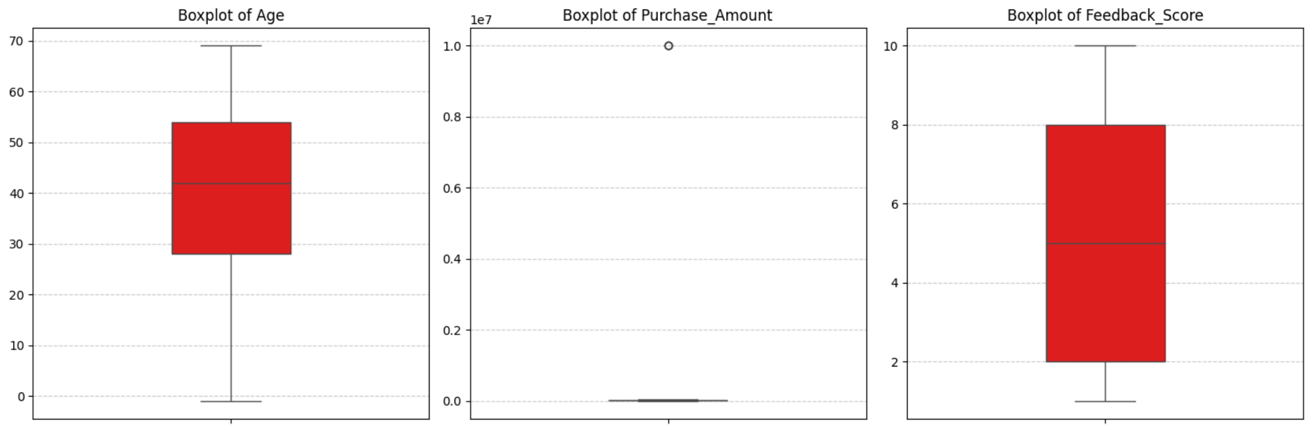
```
# Plotting the box plot to check for outliers present in the dataset
import matplotlib.pyplot as plt
import seaborn as sns

cols = ['Age', 'Purchase_Amount', 'Feedback_Score']
plt.figure(figsize = (15,5))

for i, col in enumerate (cols, 1):
  plt.subplot(1,3,i)
  sns.boxplot(y=df[col], color = 'red', width = 0.3)
  plt.title(f'Boxplot of {col}', fontsize = 12)
  plt.ylabel('')
  plt.grid(axis = 'y', linestyle = '--', alpha = 0.6)

plt.tight_layout()
plt.show()
```






```
from scipy import stats
```

```
# Removing the outliers by z-score method
import numpy as np
zscores = np.abs(stats.zscore(df[['Age', 'Purchase_Amount', 'Feedback_Score']]))
df = df[(zscores < 3).all(axis = 1)]
```

```
df
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	kolkata	2025-06-26	2025-05-17	26944.0	1
1	CUST5957	Jason Thomas	Male	51	kolkata	2021-02-17	2025-07-22	44152.0	2
2	CUST3754	Brittney Martinez	Female	62	hyderabad	2023-11-05	2024-12-08	31745.0	2
3	CUST2934	Brenda Pierce	Female	4	hyderabad	2022-03-13	2025-10-02	39674.0	1
4	CUST5683	Matthew Carroll	Female	41	chennai	2024-04-05	2024-12-15	24268.0	8
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	kolkata	2023-03-04	2025-06-30	43655.0	6
10194	CUST6146	Cody Thompson	Female	4	kolkata	2024-08-21	2024-10-26	35101.0	1
10195	CUST10767	Robert Lewis	Female	35	delhi	2020-12-08	2025-01-25	24167.0	9
10197	CUST6315	Joshua Martinez	Male	25	hyderabad	2022-02-15	2025-01-11	43832.0	2
10198	CUST4812	Sarah Miller	Female	55	kolkata	2021-03-16	2025-05-14	18827.0	10

8093 rows × 12 columns

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	kolkata	2025-06-26	2025-05-17	26944.0	1
1	CUST5957	Jason Thomas	Male	51	kolkata	2021-02-17	2025-07-22	44152.0	2
2	CUST3754	Brittney Martinez	Female	62	hyderabad	2023-11-05	2024-12-08	31745.0	2
3	CUST2934	Brenda Pierce	Female	4	hyderabad	2022-03-13	2025-10-02	39674.0	1
4	CUST5683	Matthew Carroll	Female	41	chennai	2024-04-05	2024-12-15	24268.0	8
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	kolkata	2023-03-04	2025-06-30	43655.0	6
10194	CUST6146	Cody Thompson	Female	4	kolkata	2024-08-21	2024-10-26	35101.0	1
10195	CUST10767	Robert Lewis	Female	35	delhi	2020-12-08	2025-01-25	24167.0	9
10197	CUST6315	Joshua Martinez	Male	25	hyderabad	2022-02-15	2025-01-11	43832.0	2
10198	CUST4812	Sarah Miller	Female	55	kolkata	2021-03-16	2025-05-14	18827.0	10

8093 rows × 12 columns

Next steps: Generate code with df | New interactive sheet | Generate code with df | New interactive sheet

```
df.shape
```

```
(8093, 12)
```

```
(8093, 12)
```


```
df['City'] = df['City'].str.capitalize()
```

```
df['City']
```

	City
0	Kolkata
1	Kolkata
2	Hyderabad
3	Hyderabad
4	Chennai
...	...
10193	Kolkata
10194	Kolkata
10195	Delhi
10197	Hyderabad
10198	Kolkata

8093 rows × 1 columns

dtype: object

	City
0	Kolkata
1	Kolkata
2	Hyderabad
3	Hyderabad
4	Chennai
...	...
10193	Kolkata
10194	Kolkata
10195	Delhi
10197	Hyderabad
10198	Kolkata

8093 rows × 1 columns

dtype: object

```
# Just formatting so that dataset looks clean
df['Country'] = df['Country'].str.capitalize()
```

```
df
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	Kolkata	2025-06-26	2025-05-17	26944.0	1
1	CUST5957	Jason Thomas	Male	51	Kolkata	2021-02-17	2025-07-22	44152.0	2
2	CUST3754	Brittney Martinez	Female	62	Hyderabad	2023-11-05	2024-12-08	31745.0	2
3	CUST2934	Brenda Pierce	Female	4	Hyderabad	2022-03-13	2025-10-02	39674.0	1
4	CUST5683	Matthew Carroll	Female	41	Chennai	2024-04-05	2024-12-15	24268.0	8
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	Kolkata	2023-03-04	2025-06-30	43655.0	6
10194	CUST6146	Cody Thompson	Female	4	Kolkata	2024-08-21	2024-10-26	35101.0	1
10195	CUST10767	Robert Lewis	Female	35	Delhi	2020-12-08	2025-01-25	24167.0	9
10197	CUST6315	Joshua Martinez	Male	25	Hyderabad	2022-02-15	2025-01-11	43832.0	2
10198	CUST4812	Sarah Miller	Female	55	Kolkata	2021-03-16	2025-05-14	18827.0	10

8093 rows × 12 columns

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score
0	CUST4371	Paul Wilson	Male	52	Kolkata	2025-06-26	2025-05-17	26944.0	1
1	CUST5957	Jason Thomas	Male	51	Kolkata	2021-02-17	2025-07-22	44152.0	2
2	CUST3754	Brittney Martinez	Female	62	Hyderabad	2023-11-05	2024-12-08	31745.0	2
3	CUST2934	Brenda Pierce	Female	4	Hyderabad	2022-03-13	2025-10-02	39674.0	1
4	CUST5683	Matthew Carroll	Female	41	Chennai	2024-04-05	2024-12-15	24268.0	8
...	...	...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	Female	26	Kolkata	2023-03-04	2025-06-30	43655.0	6
10194	CUST6146	Cody Thompson	Female	4	Kolkata	2024-08-21	2024-10-26	35101.0	1
10195	CUST10767	Robert Lewis	Female	35	Delhi	2020-12-08	2025-01-25	24167.0	9
10197	CUST6315	Joshua Martinez	Male	25	Hyderabad	2022-02-15	2025-01-11	43832.0	2
10198	CUST4812	Sarah Miller	Female	55	Kolkata	2021-03-16	2025-05-14	18827.0	10

8093 rows × 12 columns

Next steps: Generate code with df | New interactive sheet | Generate code with df | New interactive sheet

```
df[df['Age'] == -1]
```